21 — Solid Track: Sorbent Selection & Synthesis

v1.0 — material logic, both synthesis routes, and QC gates

Function: Select and produce the desiccant that makes doc 20 work: capture at ~80% RH ambient, release against a *humid* purge at the lowest practical temperature, benign chemistry in breathing air, distilled-clean condensate, marine-grade durability, and DIY procurability.

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1. Selection framework — the isotherm step

The decisive property is the **shape and position of the water-uptake isotherm**, not peak capacity. Type V (S-shaped) sorbents have a sharp uptake step at a characteristic RH: nearly empty below it, nearly full above it. Capture at 80% RH is trivial for all candidates; **regeneration is the whole game**, and the step position sets the regeneration temperature.

> **Rule:** pick the material whose step sits **as high as possible while still below the minimum capture humidity** — this minimizes the temperature swing needed to release water against humid surroundings.

The inverse-problem insight: most published sorbents are tuned for *arid* harvesting (step ≤30% RH tuned for capture), where capture is hard and release easy. This application is the mirror image — at 80% RH the sorbent gorges itself, and release against near-saturated surroundings is the entire challenge. Design priorities flip: optimize for cheap, fast, **complete desorption**, and engineer against **deliquescence** at 80%+ RH.

2. Candidates and verdict

Option	Step / regen (humid purge)	Working Δq (g/g)	Durability	Breathing air / marine	Procurement	Verdict
Aluminum fumarate (AlFu / A520 / MIL-53(Al)-FA)	step ~25–30% RH → **~60–65 °C**	0.2–0.3 (dry-purge basis — F3 note §6)	excellent; 4,500 cycles unchanged, sibling CAU-10-H to 10,000; fails only mechanically	benign, inert, clean condensate	DIY LAG + commercial validation lot	**Selected**
Silica gel (incl. commercial wheel)	gradual, ~80–120 °C	0.2–0.4	10^5+ cycles, marine-proven	inert, clean	trivial; turnkey wheels	**Strongest rival / benchmark (T8)** — loses only on regen grade + DCHX elegance
Molecular sieve 13X / activated alumina	~150–300 °C	0.2–0.25	bulletproof	inert	trivial, bulk	High-grade-heat-only → forfeits the low-temp comfort island. Complement, not primary
AQSOA-Z05 (AlPO)	step ~50–60% RH, low regen	0.15–0.2	100k cycles	inert	small lots hard; ~2× inventory	Viable, not superior
CAU-23 / CAU-10-H	~60 °C, step ~30% RH	≈ AlFu	≈ AlFu class	benign	not commercial	**Upgrade path** (deferred until AlFu proven)
MOF-303 / MIL-160	step 5–20% RH (needs drier/hotter purge)	—	good	benign	hard	Wrong step direction — worse here
LiCl/$CaCl_2$ liquid desiccant	50–70 °C	high	self-renewing	corrosive aerosol carryover **absent demonstrated aerosol control**	commercial LDAC / this repo's liquid track	**Rejected for cabin air PENDING test B** — the liquid track's eliminator + demister + coupon chain (doc 11 §3) is the arbiter; if B passes, the tracks layer rather than compete (X3, doc 40)
Salt-composite / hydrogel sorbents	low regen	0.35–0.7	swell/leach/biofouling	conditional at best	DIY-hard	Rejected — the failure modes are exactly the field ones

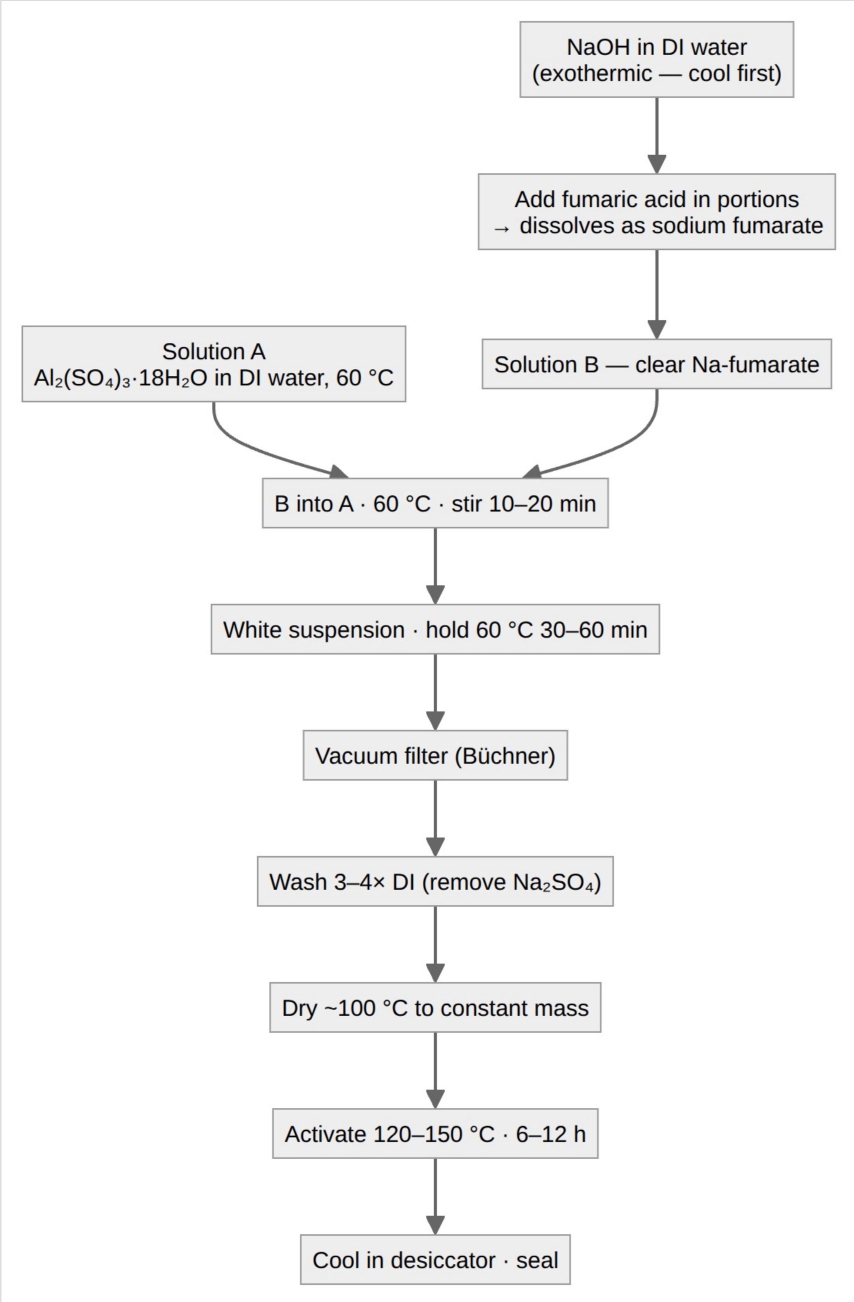
Why AlFu wins this constraint set: DIY-synthesizable from benign, residential-shippable reagents; inert crystalline framework (no biocide chemistry, no deliquescence engineering, no condensate polish); potable-clean condensate; lowest practical regeneration among durable, makeable-or-buyable solids. **Caveat (F2):** with solar-direct regeneration dead at the peak point, AlFu's edge over silica gel narrows to chemistry, condensate quality, DIY-ability, and a wider usable slice of the heat cascade — still a win, but benchmark against a commercial silica-wheel quote (T8) and hold a commercial Basolite A520 validation lot (T9) so coating work never blocks on synthesis.

3. Route A — aqueous precipitation (turnkey chemistry)

The framework is **[Al(OH)(O_2C–CH=CH–CO_2)]**, 1 Al : 1 fumarate. Aqueous metathesis:

1. **Linker activation:** `H₂Fum + 2 NaOH → Na₂Fum + 2 H₂O`
2. **Framework precipitation at ~60 °C:** `Al³⁺ + Fum²⁻ + H₂O → Al(OH)(Fum)↓ + H⁺` (sulfate spectates, leaves as benign Na_2SO_4, washed out)

No toxic gas evolved; hazards are NaOH corrosivity and mild Al-salt acidity.



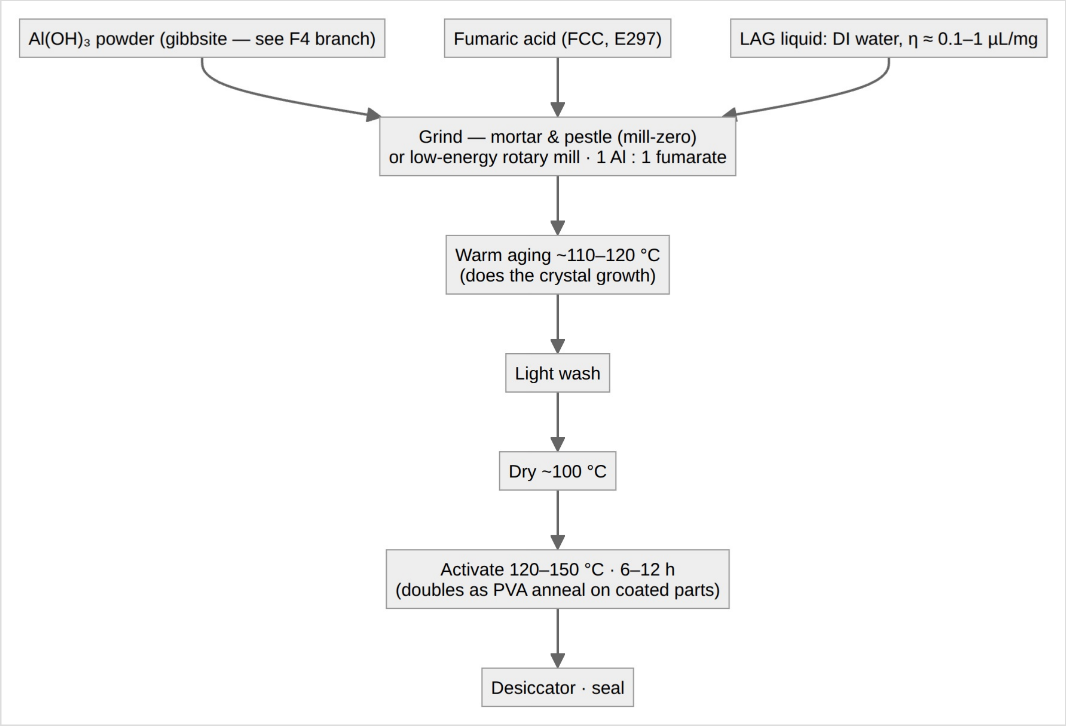
Reagent quantities (scale linearly; verify hydrate assay per lot COA):

Reagent	Validation (~19 g MOF)	Bench (~75 g MOF)	Full charge (~2.5 kg, indicative)
$Al_2(SO_4)_3 \cdot 18H_2O$ (MW 666.43)	41.7 g	**166.6 g**	~5.5 kg
Fumaric acid (MW 116.07)	14.5 g	**58.0 g**	~1.9 kg
NaOH (MW 40.00)	10.0 g	**40.0 g**	~1.3 kg
DI water total	~190 mL	**~750 mL**	linear

Mole check (bench): 0.50 mol Al : 0.50 mol fumarate : 1.00 mol NaOH = 1 : 1 : 2.

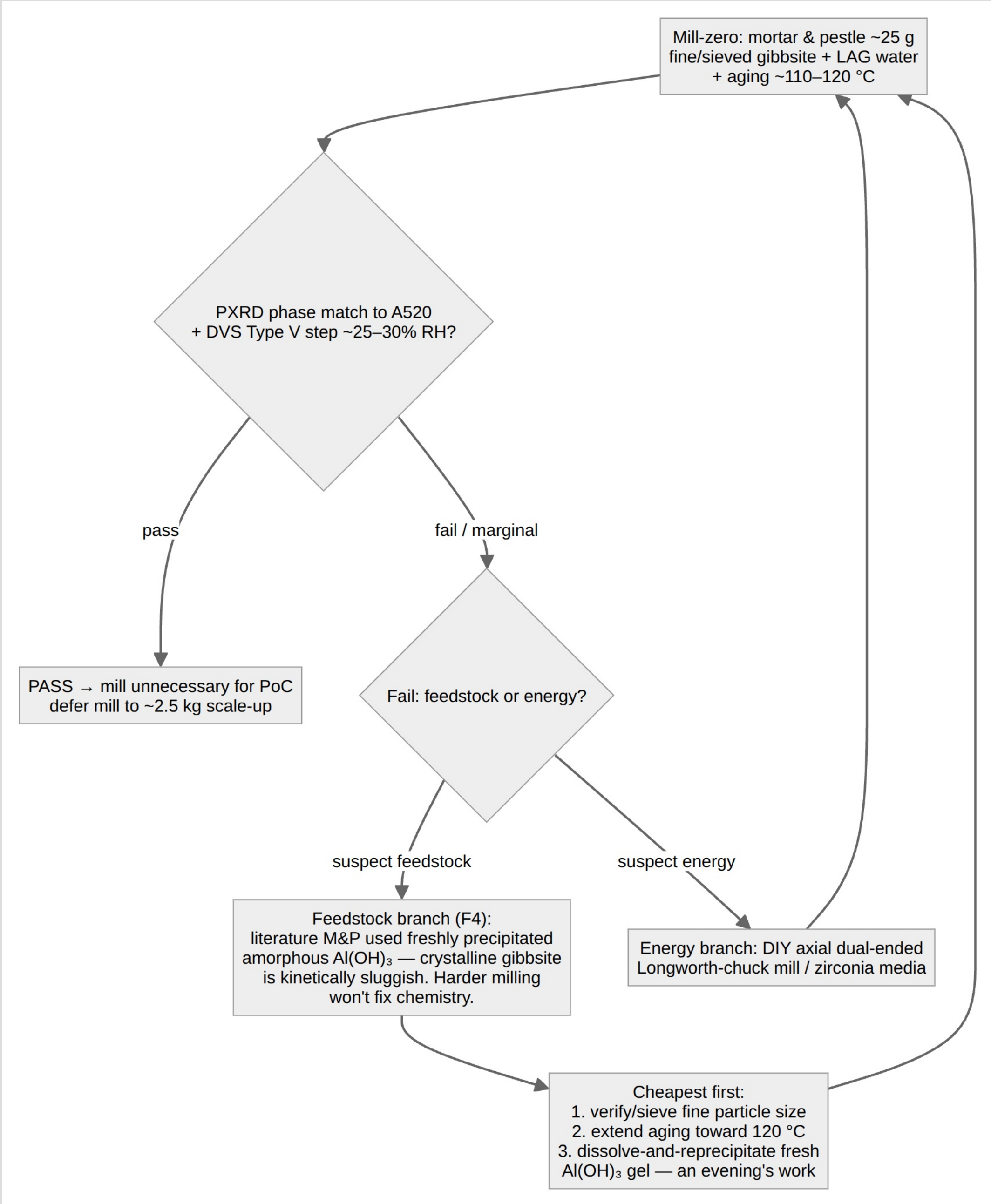
4. Route B — LAG mechanochemical (selected PoC path)

Liquid-assisted grinding: $Al(OH)_3$ + fumaric acid + catalytic DI water, **no base, no sulfate waste, minimal wash**. AlFu is the mildest known MOF mechanochemistry case — made by **mortar and pestle plus oven aging** — so energy is *not* the constraint; the **aging step does much of the crystal growth**.



Developmental parameters (open until frozen against §6 QC): stoichiometry 1:1, no separate base; LAG water at $\eta \approx 0.1$–1 µL/mg; optional ILAG salt modulator; mill speed/time and ball-to-powder ~10:1–30:1 (mill path only); two-stage mill→age is the working hypothesis. Expected surface area ~790–1,100 m^2/g (vs ~1,135 aqueous) — acceptable, since the binding gates are phase purity and the DVS step, not BET.

Mill-zero gate and the feedstock-reactivity branch (F4):



Run the reprecipitated-gel variant **before** building the mill. The mill returns at scale-up regardless (hand-grinding does not scale to ~2.5 kg).

Sourcing pattern (generalizable): non-hazmat, residential-shippable reagents ($Al(OH)_3$, fumaric acid, PVA); local industrial distributors for bulk alumina hydrate; a commercial Basolite A520 validation lot (~25–100 g) in parallel — **buy-to-validate, make-to-scale**.

5. Aluminum-source note

Activated alumina **cannot** be converted into an AlFu precursor by adsorbing humidity — vapor uptake is reversible surface rehydroxylation, not bulk conversion to gibbsite. The aluminum source remains gibbsite (fresh-precipitated preferred, per F4) or Al salts.

6. QC acceptance criteria — defines "good" AlFu

Run 1–2 on every batch; 3–5 on the first batch of each new reagent lot or process change. Freeze the recipe once a batch passes all.

#	Gate	Criterion	Notes
1	Appearance	free-flowing off-white/white powder	grey/yellow/tan → suspect Fe or organic contamination
2	Yield	activated dry mass ≥ 90% of theoretical (~158 g/mol per Al)	
3	**PXRD**	matches reference A520 / MIL-53(Al)-FA, sharp reflections	compare against a simulated-CIF or literature pattern directly; never rely on memorized peak positions
4	BET (N_2, 77 K)	aqueous ≥ 1,000 m²/g (target ~1,100–1,135) · grinding ≥ 750–800	low BET + correct PXRD usually = incomplete activation → re-activate, re-run
5	**DVS**	**Type V isotherm, step ~25–30% RH, working uptake ~0.2–0.3 g/g, reversible**	the real pass/fail gate together with PXRD
6	TGA (optional)	one clean dehydration event before framework decomposition	activation completeness

F3 annotation (mandatory on all sizing tables): 0.2–0.3 g/g is a **dry-purge full-swing** capacity. Effective swing = uptake − residual(regeneration at 60–65 °C against ~25 g/kg purge); a plausible 0.05–0.10 g/g residual cuts effective Δq to ~0.15–0.2 g/g (+25–50% inventory), compounding F1. M2 extracts **effective Δq under representative purge** as a first-class output.

Characterization is outsourced (PXRD + DVS at a university fee-for-service facility); a submission pack (benign, non-hazmat hazard summary + requested runs) accompanies each batch. DIY instrumentation covers everything else.

Version history

- **v1.0** — New lineage from archived 02: liquid-desiccant verdict re-scoped "PENDING test B" per X3; synthesis routes, F4 branch, and QC gates carried verbatim in substance; CO_2-sorbent MOF upgrade path noted in doc 12 §6.